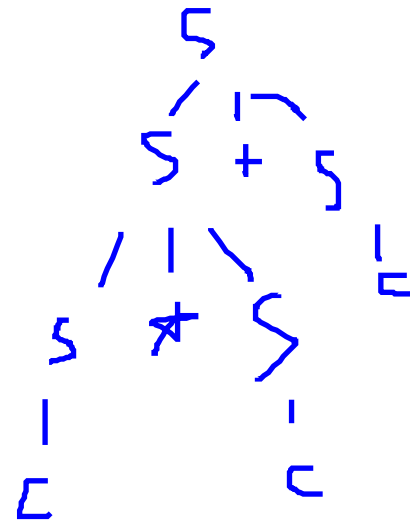
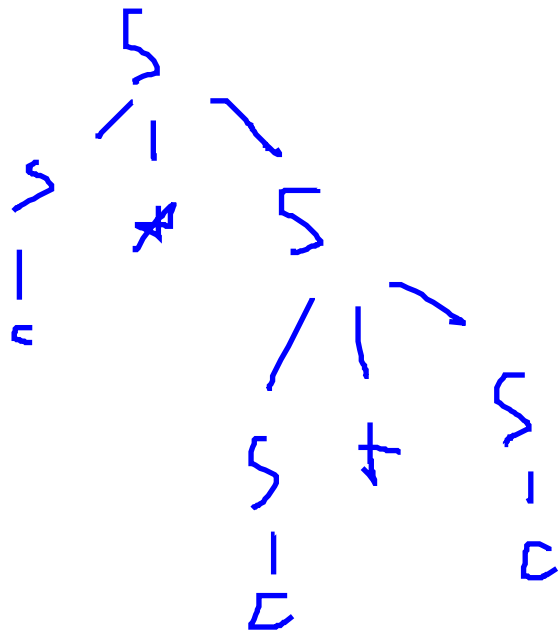


Theory of Computation

Exercise 8: (Context-free grammar part 1)

1. Prove that the following grammar is ambiguous.

$$S \rightarrow S + S \mid S - S \mid S * S \mid S / S \mid c$$



2. Find CFG for the language L.

$$L = \{ a^i b^j : i \leq j \}$$

λ, b, ab, abb

$$S \rightarrow AB$$

$$A \rightarrow aABb \mid \lambda$$

$$B \rightarrow Bb \mid \lambda$$

$$\begin{array}{l} S \rightarrow aSb \mid B \mid \lambda \\ B \rightarrow bB \mid \lambda \end{array}$$

$$S \rightarrow aSb \mid Sb \mid \lambda$$

*3. Find the language of the following grammar.

(Homework 6)

$$G: S \rightarrow aA \mid bA \mid a \mid b$$

$$A \rightarrow aS \mid bS$$

$a, b, aab, aba, abb,$
 $bba, bab, bba, bbb,$

$$L = \{w \text{ in } \{a,b\}^* : |w| \text{ is odd}\}$$